

Our Solar System

A plain-language guide to the Sun and the eight planets.

NOTE A synthetic, simplified document for a software demo. Facts are kept short and general; it is not a textbook. Citations are illustrative.

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1. The Sun

Our star — the centre of everything

BIG IDEA The Sun holds about **99.8%** of all the mass in the solar system; everything else orbits it [1].

In plain words

The Sun is a giant ball of hot gas that makes light and heat by fusing hydrogen into helium in its core. Its gravity holds all the planets in their orbits. Without it there would be no daylight, no seasons, and no life [1].

DID YOU KNOW If the whole solar system were a stadium, our star would be almost the entire stadium — the planets are just specks in the seats.

2. Mercury

Inner rocky planet — closest to the Sun

BIG IDEA A year on Mercury is only **88 Earth days** — it races around the Sun faster than any other planet [2].

In plain words

Mercury is the smallest planet and the nearest to the Sun. It is a scorched, cratered ball of rock with almost no air, so it bakes in sunlight and freezes in shadow [2].

DID YOU KNOW With barely any atmosphere to trap heat, the same spot can be hundreds of degrees by day and bitterly cold at night.

3. Venus

Inner rocky planet — Earth's hot twin

BIG IDEA Venus is the **hottest planet** (about 465°C) because a thick carbon-dioxide blanket traps heat — a runaway greenhouse [3].

In plain words

Venus is almost the same size as Earth, which is why it is called Earth's twin. But its heavy atmosphere makes it scorching, and oddly it spins backwards — a day there is longer than its year [3].

DID YOU KNOW Sometimes called the 'morning star', Venus is the brightest natural object in our night sky after the Moon.

4. Earth

Inner rocky planet — the living one

BIG IDEA Earth is the only place we know of with **life**, liquid water, and air we can breathe [4].

In plain words

About 71% of Earth's surface is covered by water. A protective atmosphere and a comfortable distance from the Sun keep temperatures right for liquid oceans and living things [4].

DID YOU KNOW Seen from space Earth looks like a 'pale blue dot' — the blue is all that ocean water.

5. Mars

Inner rocky planet — the Red Planet

BIG IDEA Mars looks **rusty red** because its soil is full of iron oxide — literally rust [5].

In plain words

Mars is a cold desert world with a thin atmosphere. It has the tallest volcano in the solar system (Olympus Mons) and dried-up channels that suggest rivers once flowed there [5].

DID YOU KNOW Because it may once have had liquid water, Mars is the top target in the search for past life beyond Earth.

6. Jupiter

Outer gas giant — the biggest planet

BIG IDEA Jupiter is so large it could swallow **over 1,300 Earths**; it is more than 300 times Earth's mass [6].

In plain words

Jupiter is a gas giant with no solid surface, just deep swirling clouds. Its Great Red Spot is a storm bigger than our whole planet that has raged for centuries, and it has dozens of moons [6].

DID YOU KNOW Jupiter acts a bit like a cosmic shield, its huge gravity pulling in comets that might otherwise hit the inner planets.

7. Saturn

Outer gas giant — the ringed planet

BIG IDEA Saturn is so **light for its size** that, given a big enough bathtub, it would float in water [7].

In plain words

Saturn is the second-largest planet, famous for its bright rings made of countless chunks of ice and rock. Like Jupiter it is mostly gas, which is why it is the least dense planet [7].

DID YOU KNOW The rings are dazzling but thin — only about tens of metres thick in places, like a sheet of paper the width of a city.

8. Uranus & Neptune

Outer ice giants — the windy far worlds

BIG IDEA Neptune has the **fastest winds** in the solar system, reaching about 2,000 km/h [8].

In plain words

Uranus and Neptune are 'ice giants', made largely of icy material, and both look blue. Uranus is tipped almost completely on its side, so it rolls around the Sun, while distant Neptune is a deep, storm-blown blue [8].

DID YOU KNOW They are so far away that sunlight there is hundreds of times fainter than on Earth — a permanent deep twilight.

References

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